

content of image, hence net Q is also zero. By 1st law of thermodynamics, W is also zero.

- 14 The air in an aircraft when travelling has the following energy as listed in the table below.

Type of energy	As a result of	Amount of energy / MJ
Kinetic energy	motion of the aircraft	8
Kinetic energy	random movement of the air molecules	30
Potential energy	altitude of the aircraft	75
Potential energy	intermolecular attraction between the air molecules	-3

What is the internal energy of the air in the aircraft?

- A** 27 MJ **B** 33 MJ **C** 35 MJ **D** 110 MJ

Ans: A

Internal energy only includes kinetic energy due to random movement of the air molecules and potential energy due to intermolecular attraction between the air molecules.

- 15 When sound travels through air, the particles oscillate. A graph of displacement against time for